

CLOUD COMPUTING PROJECT REPORT

Title: Launch and Use a Cloud Virtual Machine

Name: Dilan Mathias

Course: Cloud Computing

Date: 02/05/2026

1. Introduction

Cloud computing is a modern technology that allows users to access computing resources such as servers, storage, and applications over the internet instead of relying on local machines. It eliminates the need for physical hardware and enables users to work from anywhere using remote systems hosted in data centres.

In this project, a cloud-based virtual machine was created using Amazon Web Services through its Amazon EC2 service. The virtual machine acts like a real computer, allowing installation of software, execution of commands, and hosting of applications.

The main objective of this project is to understand the practical working of cloud infrastructure by launching a virtual machine, connecting to it remotely using SSH, installing a web server, and accessing it through a web browser. This provides hands-on experience of how real-world cloud systems operate.

2. Objective

The objective of this project is to understand the practical implementation of cloud computing by creating and managing a virtual machine in a cloud environment. This includes learning how to configure system resources such as CPU, memory, and storage, and accessing the virtual machine remotely over the internet.

The project also aims to provide hands-on experience in connecting to the virtual machine using SSH, installing and running a web server, and accessing the hosted web page through a browser. Through this, the project demonstrates how real-world cloud infrastructure operates and how applications can be deployed and managed remotely.

- Create a virtual machine in the cloud
- Configure system resources
- Connect remotely using SSH
- Install a web server
- Access it through a browser

3. Tools and Platform Used

Platform: Amazon Web Services (AWS)

Service: EC2

Operating System: Ubuntu Server

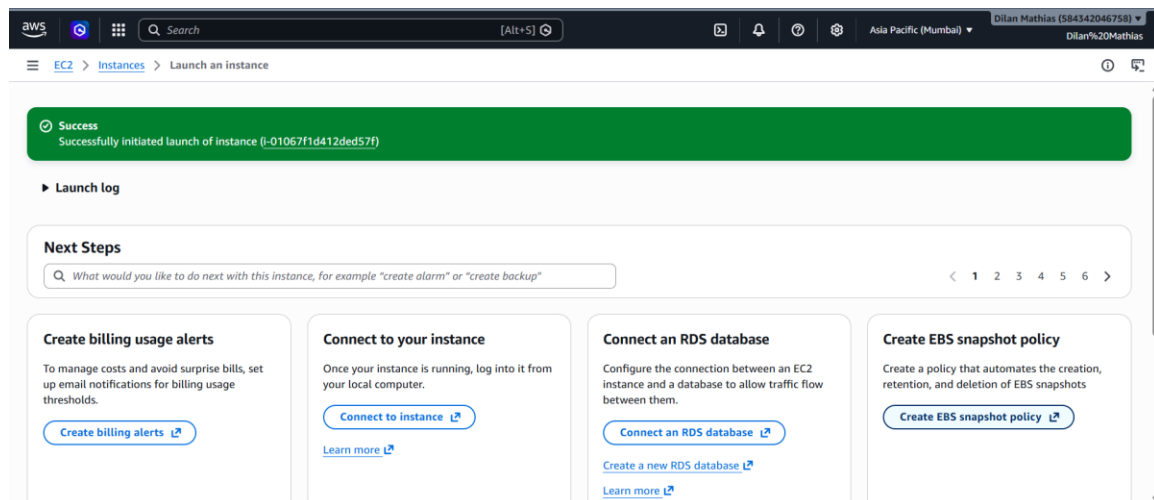
Protocol: SSH

Web Server: Apache2

4. Steps Performed

Step 1: Creating the Virtual Machine

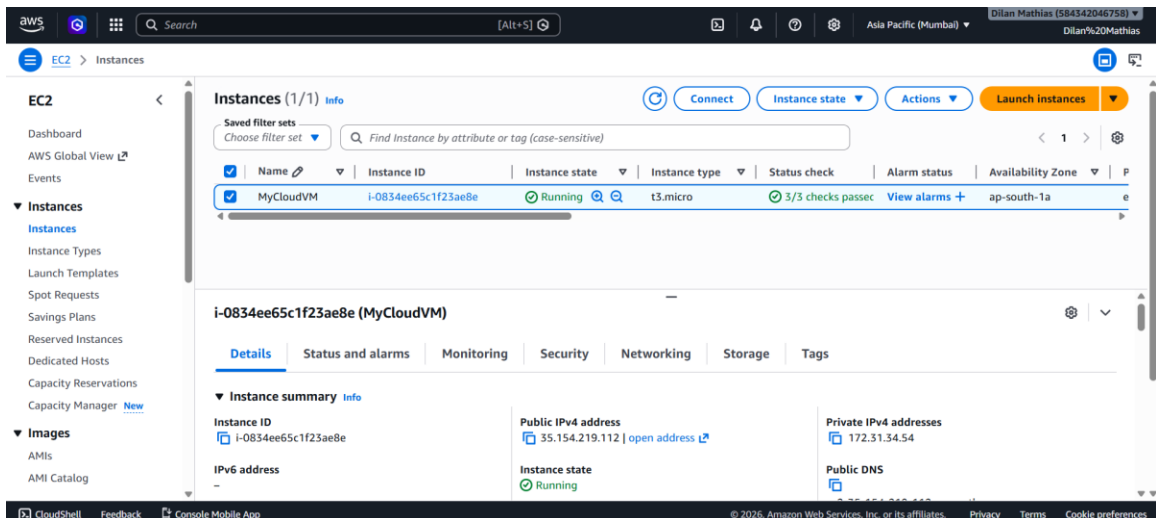
Logged into AWS, selected EC2 service, launched an instance using Ubuntu OS, selected t3.micro (free tier), configured storage and network, created key pair, and launched the instance.



VM Running

Step 2: Connecting to the Virtual Machine

Connected using EC2 Instance Connect browser-based SSH terminal using the public IP address.



Terminal Connection

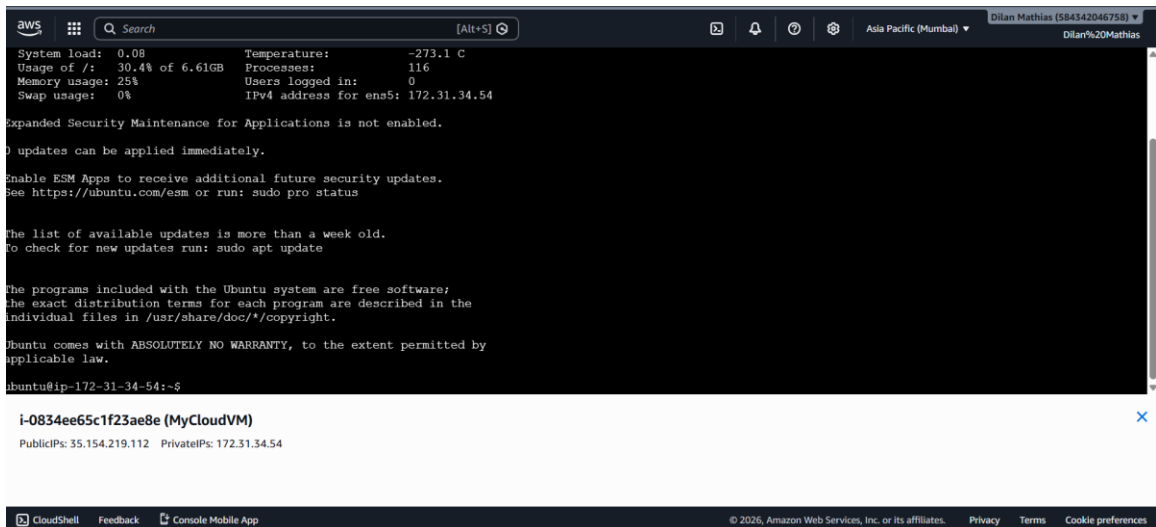
Step 3: Installing Web Server

Commands used:

```
sudo apt update
```

```
sudo apt install apache2 -y
```

```
sudo systemctl start apache2
```



Step 4: Accessing Web Page

Opened browser and entered the public IP address. The Apache default page was displayed.



Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in /usr/share/doc/apache2/README.Debian.gz**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the **apache2-doc** package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- **apache2.conf** is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- **ports.conf** is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the **mods-enabled/**, **conf-enabled/** and **sites-enabled/** directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective ***-available/** counterparts. These should be managed by using our helpers **a2enmod**, **a2dismod**, **a2ensite**, **a2dissite**, and **a2enconf**, **a2disconf** . See their respective man pages for detailed information.
- The binary is called **apache2** and is managed using **systemd**, so to start/stop the service use **systemctl start apache2** and **systemctl stop apache2**, and use **systemctl status apache2** and **journalctl -u apache2** to check status. **system** and **apache2ctl** can also be used for service management if desired. **Calling /usr/bin/apache2 directly will not work** with the default configuration.

Document Roots
By default, Ubuntu does not allow access through the web browser to <i>any</i> file apart from those located in <code>/var/www</code>, public_html directories (when enabled) and <code>/usr/share</code> (for web applications). If your site is using a web document root located elsewhere (such as in <code>/srv</code>) you may need to whitelist your document root directory in <code>/etc/apache2/apache2.conf</code>. The default Ubuntu document root is <code>/var/www/html</code>. You can make your own virtual hosts under <code>/var/www</code>.
Reporting Problems
Please use the <code>ubuntu-bug</code> tool to report bugs in the Apache2 package with Ubuntu. However, check existing bug reports before reporting a new bug. Please report bugs specific to modules (such as PHP and others) to their respective packages, not to the web server itself.

5. Results

The cloud virtual machine was successfully created and accessed remotely. The Apache web server was installed and hosted a web page accessible via browser.

6. Key Concepts Learned

- Cloud computing
- Virtual machines
- Remote access using SSH
- Web server deployment
- Networking basics

7. Conclusion

This project successfully demonstrated the practical use of cloud computing by creating and managing a virtual machine in a cloud environment. A virtual machine was launched using Amazon Web Services through the Amazon EC2 service, and it was accessed remotely using SSH.

A web server (Apache) was installed and configured on the virtual machine, and the hosted web page was successfully accessed through a browser using the public IP address. This confirmed that the server was functioning correctly and accessible over the internet.

Through this project, I gained hands-on experience in working with cloud infrastructure, remote server management, and basic web hosting. It also helped me understand how real-world cloud systems operate, where applications run on remote servers and are accessed globally via the internet.